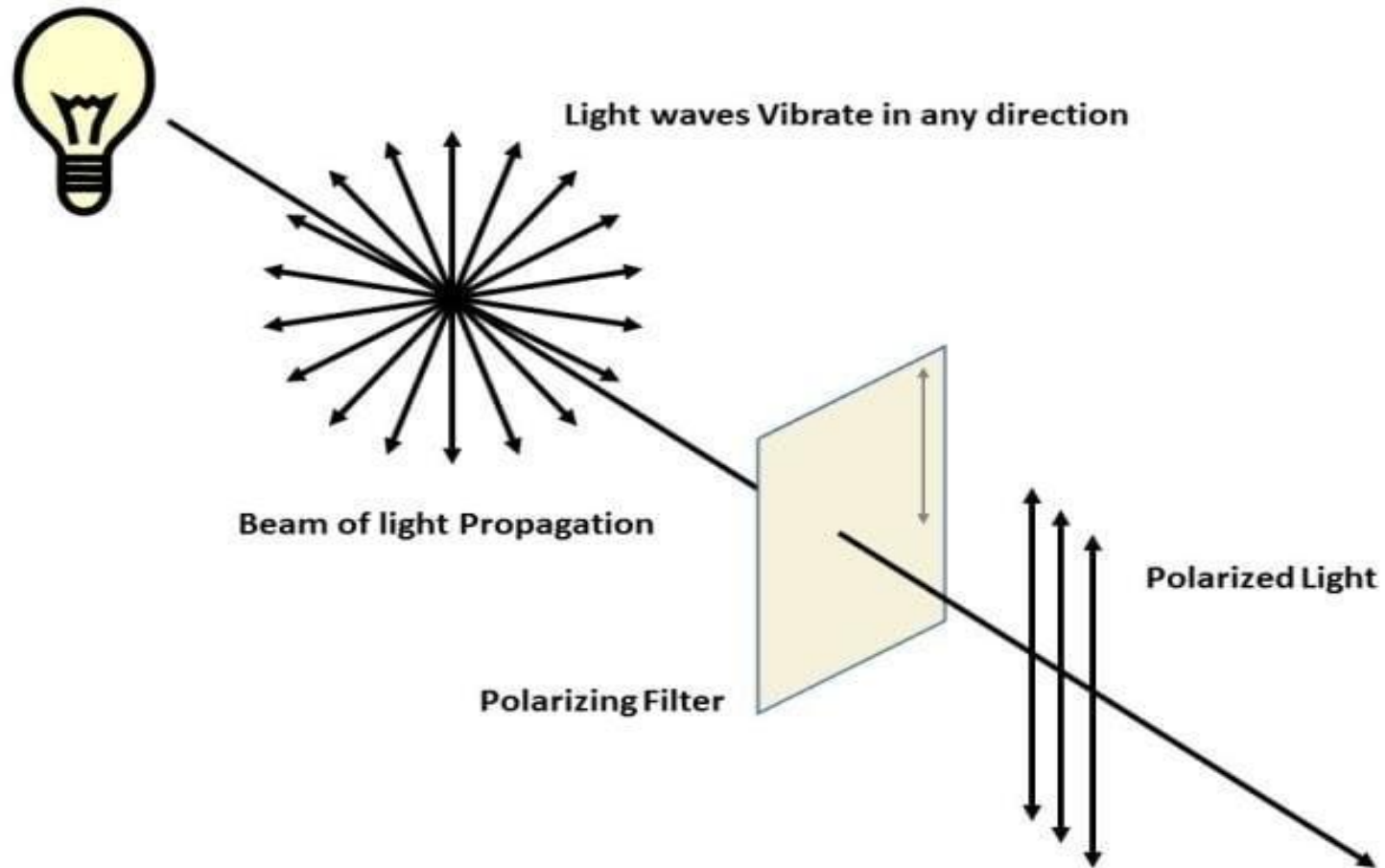


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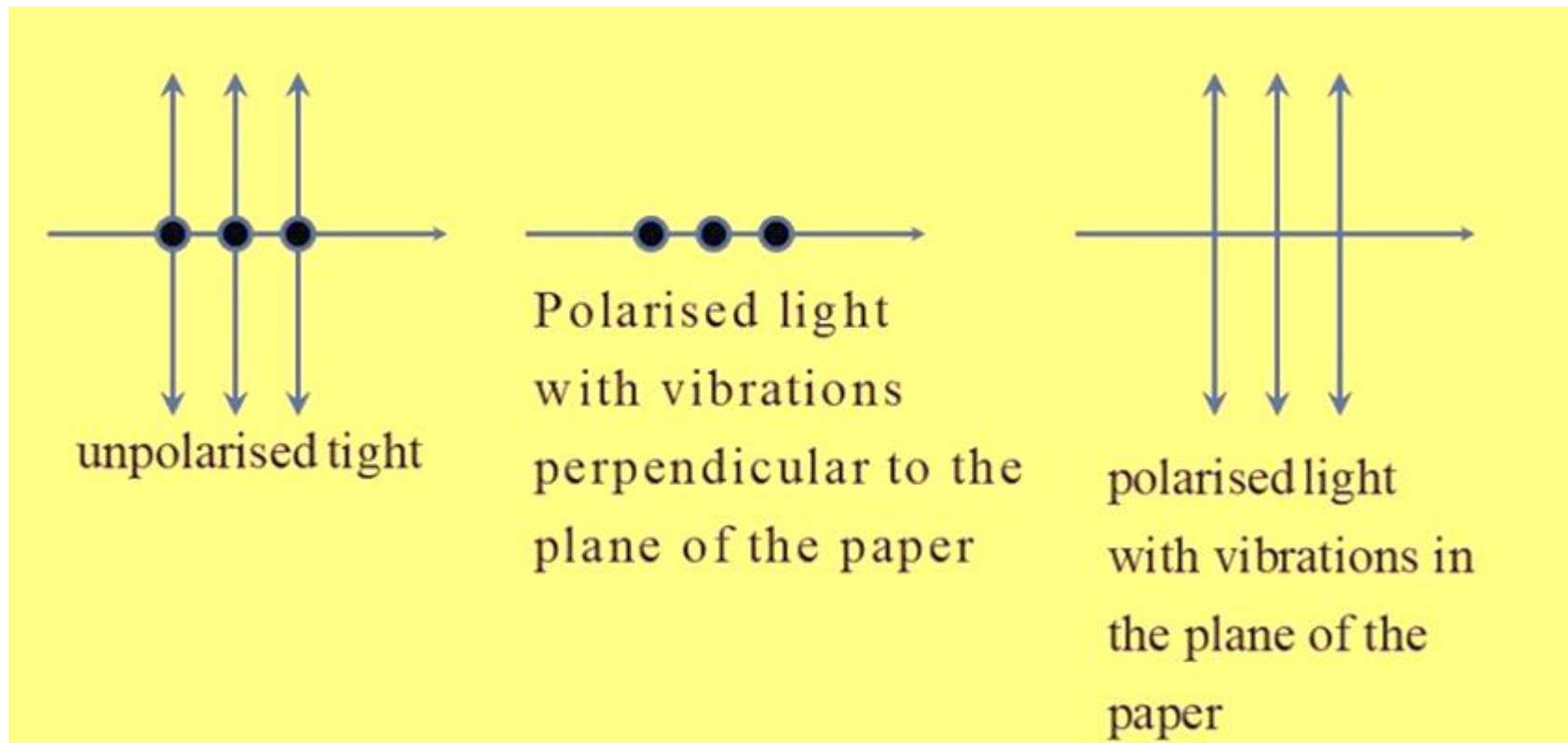
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Polarization



- ❖ The light in which the vibration of electric field vector is in all directions perpendicular to direction of propagation of light is called unpolarized light.
- ❖ The light in which the vibration of electric field vector is confined in specific direction perpendicular to propagation is called polarized light.
- ❖ The process of converting unpolarized light into polarized light is called polarization.
- ❖ Polarization is a property of transverse waves like light.
- ❖ It describes the orientation of the electric field vector and can be explained using Electromagnetic Waves
- ❖ There are different types of polarized light like linear polarized, circular polarized, elliptical polarization.



- ❖ There are different methods of polarization like: by reflection (Brewster's Law), by refraction, by selective absorption (Polaroids), by scattering.
- ❖ There are many applications of polarization like LCD screens and displays, Sunglasses, Photography filters, Optical instruments and communication

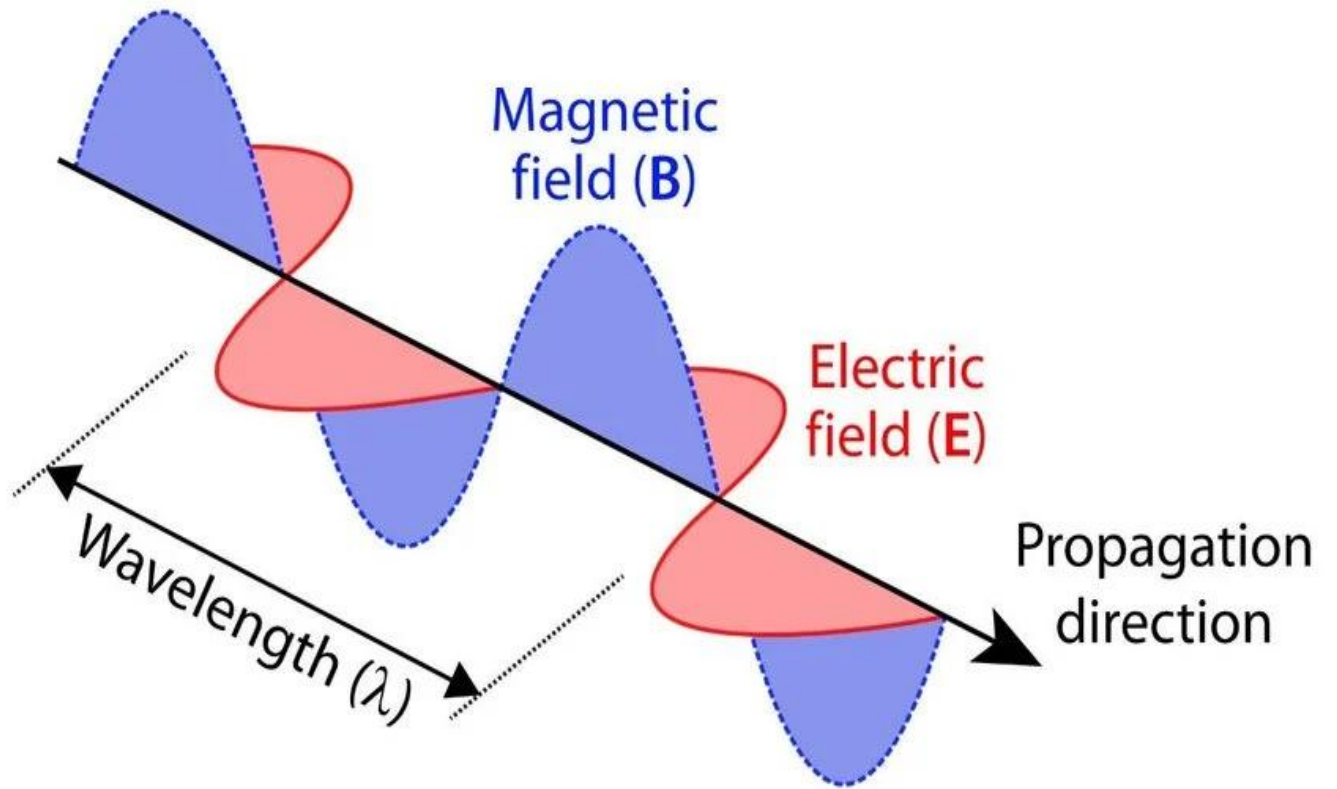
Application of electromagnetic theory (EMT) in Wi-Fi, antennas, Bluetooth

- ❖ Wi-Fi stands for Wireless Fidelity.
- ❖ This was coined by Interbrand (branding firm) for the Wi-Fi alliance to promote a catchy name.
- ❖ It is used for smart devices, internet and Internet of Things.
- ❖ It is a wireless communication technology based on transmission of electromagnetic (EM) waves.
- ❖ The operating frequency is 2.4 GHz and 5 GHz.

Electromagnetic waves are composed of oscillating electric and magnetic fields that propagate through space, typically generated by accelerating electric charges.

These fields are perpendicular to each other and the direction of wave movement.

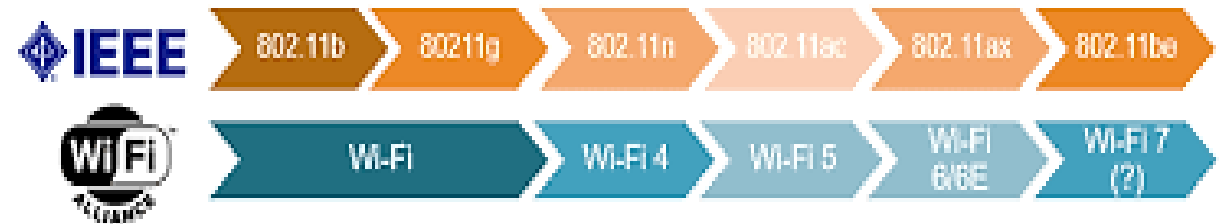
They do not require any medium for propagation.



- ❖ The antennas in Wi-Fi convert electrical signals into EM waves and vice versa.
- ❖ The antenna can be omnidirectional or directional.
- ❖ The advantages are Wireless connectivity ,Mobility and flexibility, Easy deployment and Cost-effective
- ❖ The main drawbacks are limited range, interference issues, security concerns and degradation of signal through obstacles.

Frequency Bands Used in Wi-Fi

- 2.4 GHz band:
 - Longer range
 - More interference
- 5 GHz band:
 - Higher data rates
 - Shorter range
- 6 GHz (Wi-Fi 6E):
 - Less congestion
 - High throughput

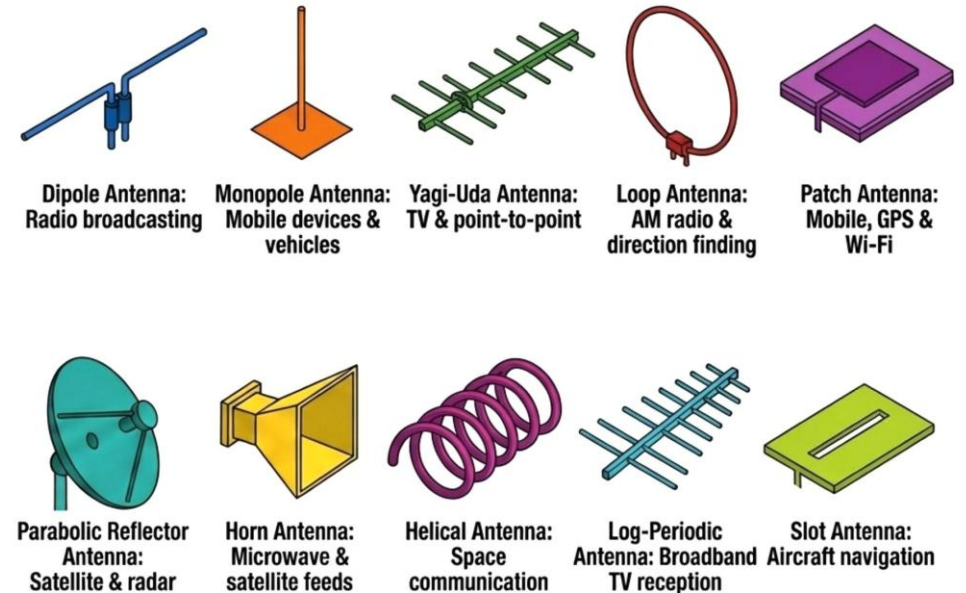


Rel. Year	1999	2007	2009	2013	2020	2023(?)
Freq. Band	2.4 GHz	2.4 GHz	2.4 + 5 GHz	5 GHz	2.4 + 5 + 6 GHz (6E)	2.4 + 5 + 6 GHz
Bandwidth	20 MHz	20 MHz	40 MHz	80 MHz, 160 MHz	80 MHz, 160 MHz	240 MHz, 320 MHz

Application of EMT in antennas

- ❖ Antennas operate based on propagation of electromagnetic waves.
- ❖ The alternating current produces time varying electromagnetic fields
- ❖ These fields get separated and propagate as electromagnetic waves
- ❖ During transmission, antenna converts electrical energy into electromagnetic waves during transmission
- ❖ During reception, antenna converts electromagnetic waves into electrical energy.
- ❖ The size of antenna is determined by wavelength commonly $\lambda/2$ (dipole) or $\lambda/4$ (monopole)

DIFFERENT TYPES OF ANTENNAS



<https://tinyurl.com/38vjwzuf>

Role of electromagnetic Theory

- ❖ The radiation pattern describe direction of signal transmission which is determined by distribution of electromagnetic field.
- ❖ The Impedance Matching reduces the reflection of signal and provide maximum power transfer.
- ❖ The gain and Directivity depend on type of interference of electromagnetic waves.
- ❖ The antennas are useful for Satellite communication, Mobile networks, Wi-Fi routers etc.

Application of electromagnetic theory (EMT) Bluetooth

- ❖ Bluetooth uses electromagnetic waves for short-range communication ($\approx 10\text{--}100\text{ m}$) with wireless connectivity.
- ❖ It Operates in the 2.4 GHz Industrial, Scientific, and Medical (ISM) band also called Frequency Hopping Spread Spectrum (FHSS)
- ❖ The data is transmitted through oscillating electric and magnetic fields.
- ❖ To reduce interference the frequencies changes rapidly.



Role of electromagnetic theory in bluetooth Performance

- ❖ **Wave Propagation Effects:** Reflection, diffraction, and absorption affect signal strength
- ❖ **Antenna Design:** Small antennas optimized for 2.4 GHz wavelength
- ❖ **Interference Management:** EM theory helps avoid overlapping with Wi-Fi and other devices
- ❖ **Power Control:** Low energy transmission reduces EM radiation and saves battery
- ❖ There are many applications like Wireless headphones & speakers, Smartwatches and fitness trackers, IoT and smart home devices